



COMP2132 ASSIGNMENT 08

OBJECTIVES

Apply animated effects to a web page.

DUE DATE

Assigned during session08, due 11:59pm the night before session09.

REQUIREMENTS

PART A: Product Carousel Control

HTML/CSS

Use the *product/* image files provided with this assignment. Design a product rotation viewer. It should display one of the product images as well as buttons for spin, reverse spin, and stopping.

SCRIPTING

When a user clicks the Spin or Reverse Spin buttons, use **requestAnimationFrame()** to swap the bicycle image from one image to the next. If the standard spin is requested show images in ascending sequence, eg: 1, 2, 3, etc up to 34 and then back to 1, 2, 3 etc to display all images over and over again. If the reverse spin is chosen, the images displayed should instead count down instead of up.

When the user clicks Stop button, cease the animation at whichever image was currently being shown through the cycle (eg: when the animation is stopped, don't go back to showing image 1).

The default spinning animation speed per frame is too fast, so add a **setTimeout()** to slow it down to one tenth of a second (100 milliseconds)

Ensure that if the user clicks any spin buttons more than once the spinning animation will not appear to speed up

PART B: Pacman Movement Controls

HTML/CSS

Use the **pacman/** images provided with this assignment. Present a box containing the static pacman and provide controls for the user to move or stop the image within the box. Ensure the box is not a fixed width, but will expand/contract in width as the viewport width is altered.

SCRIPTING

The movement controls need to provide keyboard options for moving up, down, left and right, as well as an option to stop the movement. Display information about what keys the user should use, and whichever of those options is currently active, visually distinctive CSS highlights must be applied. By default, the page should style the 'stop' option as active.

If moving, the **pac-man-fast.gif** should be used, if stopped, use the **pac-man-static.gif**. The image should always face the direction it is moving (if it is moving down, it should face down, etc).

If the user chooses to move in the same direction two consecutive times, the animation must not speed up (protect against user triggered recursive animation calls similar to exercise **session08/03-control-recursive-animation**)

If the pacman moves to any edge of the box, it should reverse direction automatically. The image should never appear to leave the box. The box and pacman's movement within it must be fully responsive to different viewport sizes (the containing box's width must change with viewport widths, and must not be a fixed width).

NOTES

Use page scope variable to track important page states such as the number of the image currently being shown, and which direction the pacman image is facing.

SUBMISSION

Before the due date, compress all assignment resources (HTML, CSS, JS, images, folders, etc) into a single **.ZIP**. Upload **.ZIP** to

learn.bcit.ca > Content > Session08 > Assignment08 Dropbox